

# VSTACK Function in Excel

## 1. What is VSTACK?

The VSTACK function joins two or more tables or ranges into one, placing each one below the previous one. It solves the problem of having similar data in separate lists, such as sales for different months or employees from different teams. A Data Analyst uses it to consolidate data without copying and pasting.

## 2. Syntax

```
=VSTACK(array1, [array2], ...)
```

| Argument    | Required? | What it accepts                                                                |
|-------------|-----------|--------------------------------------------------------------------------------|
| array1      | Required  | The first range or table to stack (for example A1:B4)                          |
| array2, ... | Optional  | More ranges or tables to place below the first one, in the order you list them |

The result has as many rows as all the arrays combined, and as many columns as the widest array. If you include the header row of the first table only, the result has just one set of headings.

## 3. Simple Example

Combine two employee lists with the same columns into one:

|   | A      | B          | C | D      | E          |
|---|--------|------------|---|--------|------------|
| 1 | Name   | Department |   | Name   | Department |
| 2 | Ayesha | Marketing  |   | Ayesha | Marketing  |
| 3 | Rahim  | Sales      |   | Rahim  | Sales      |
| 4 | Karim  | IT         |   | Karim  | IT         |
| 5 | Name   | Department |   | Lina   | HR         |
| 6 | Lina   | HR         |   | Sami   | Finance    |
| 7 | Sami   | Finance    |   | Nusrat | Operations |
| 8 | Nusrat | Operations |   |        |            |

Formula in D1 only: `=VSTACK(A1:B4, A6:B8)`  
The first list is A1:B4 (with its header). The second list is A6:B8 (data only, without its header in row 5). Excel places the second list below the first. **Result: 1 header row and 6 employees in D1 to E7**

## 4. Another Example

Combine two stock lists. The second list has no Price column:

|   | A        | B   | C     | D | E        | F   | G     |
|---|----------|-----|-------|---|----------|-----|-------|
| 1 | Item     | Qty | Price |   | Item     | Qty | Price |
| 2 | Mouse    | 40  | 25    |   | Mouse    | 40  | 25    |
| 3 | Keyboard | 25  | 45    |   | Keyboard | 25  | 45    |
| 4 |          |     |       |   | Monitor  | 15  | #N/A  |
| 5 | Item     | Qty |       |   | Headset  | 30  | #N/A  |
| 6 | Monitor  | 15  |       |   |          |     |       |
| 7 | Headset  | 30  |       |   |          |     |       |

Formula in E1 only: `=VSTACK(A1:C3, A6:B7)`  
The result is as wide as the widest list (3 columns). The second list has only 2 columns, so Excel fills the missing Price cells with #N/A. To replace them, wrap the formula in IFERROR:  
`=IFERROR(VSTACK(A1:C3, A6:B7), "")`.

## 5. When to Use It

- Combine monthly or quarterly sales lists into one table
- Merge lists from different sheets, such as one sheet per region or team
- Build a master list of customers, employees, or products
- Consolidate reports from several departments

## 6. Important Things to Remember

- VSTACK places the arrays one below the other, in the order you write them.
- VSTACK is a dynamic array function. One formula fills many cells, so the cells where the result spills must be empty, or Excel shows `#SPILL!`.
- If an array has fewer columns than the widest one, the missing cells show `#N/A`.
- Include the header row from the first table only, or the headings will repeat in the middle of the result.
- The result updates when the source data changes, and the sources can be on other sheets.
- VSTACK is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

## 7. Common Mistakes

**Mistake:** Including the header row of every table, so headings appear again in the middle of the result. **Fix:** Select the header only in the first table.

**Mistake:** Stacking tables whose columns are in a different order, so values land under the wrong headings. **Fix:** Check that the columns match in the same order before stacking.

**Mistake:** Seeing `#N/A` because the tables have different numbers of columns. **Fix:** Make the structure match, or wrap the formula in IFERROR.

**Mistake:** Getting `#SPILL!` because cells below or beside the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

## 8. Data Analyst Tip

Combine VSTACK with other functions, for example `=SORT(VSTACK(Jan!A2:C20, Feb!A2:C20))` to merge two monthly sheets and sort the result. Select only rows that contain data, because blank rows appear as 0 in the result.